

CAN BMS_BB_PKG V1.0

1. Historical changes

2. Overview

This agreement specifies the communication protocol between the BMS and other nodes in the automotive CAN network.

3. Physical interface

Physical interface This protocol adopts the CAN2.0A/B standard. The communication baud rate is 250kbps.

For multi-byte data during data transmission of this protocol, unless otherwise specified, the low byte will come first and the high byte will come last (**little endian**).

4. Parameter group number

No	Name	describe	Frame format	ID	Sender	Message cycle
1	BATT_ST1	Battery status information 1	Standard frame	0x02F4	BMS	20ms
2	CELL_VOLT	Cell voltage	Standard frame	0x04F4	BMS	100ms
3	CELL_TEMP	Battery temperature	Standard frame	0x05F4	BMS	500ms
4	ALM_INFO	Alarm message	Standard frame	0x07F4	BMS	100ms
5	BATT_ST2	Battery status information 2	Extended frames	0x18F128F4	BMS	100ms
6	Ctrl_INFO	Control information	Extended frames	0x18F0F428	Peripherals	Pending
7	ChgRequest_INFO	Battery info request charger	Extended frames	0x18F0F472 0x1806E5F4	BMS	1000ms
8	BmsSwSta	Switch status	Extended frames	0x18F528F4	BMS	500ms
9-17	CELL_VOLT_FULL	Cell voltage full	Extended frames	0x18E028F4 - 0x18E728F4	BMS	1000ms
18	CELL_TEMP_FULL	Cell temp full	Extended frames	0x18F228F4	BMS	1000ms

5. Message definition

5.1 (BATT_ST) ID: 0x02F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	BattVolt	0	16	0~1000	0.1	0	V	Total battery voltage
2	BattCurr	16	16	-400~1000	0.1	-400	A	Total current of the battery pack
3	SOC	32	8	0~100	1	0	%	Remaining capacity

5.2 (CELL_VOLT) ID: 0x04F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	MaxCellVolt	0	16	0~5000	1	0	mV	Maximum single unit voltage
2	MaxCvNO	16	8	1~250	1	1		The highest single position
3	MinCellVolt	24	16	0~5000	1	0	mV	Minimum single unit voltage
4	MinCvNO	40	8	1~250	1	1		The lowest single position

5.3 (CELL_TEMP) ID: 0x05F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	MaxCellTemp	0	8	-50~200	1	-50	°C	Maximum battery cell temperature
2	MaxCtNO	8	8	1~125	1	1		Maximum temperature position
3	MinCellTemp	16	8	-50~200	1	-50	°C	Minimum battery cell temperature
4	MinCtNO	24	8	1~125	1	1		Minimum temperature position
5	AvrgCellTemp	32	8	-50~200	1	-50	°C	Average battery cell temperature

5.4 (ALM_INFO) ID: 0x07F4

The alarm information is sent by event triggering. When there is an alarm, the BMS periodically sends the message, and if there is no alarm information, it will not be sent. When multiple alarms occur simultaneously, the instrument

The interface will display the alarm number cycle, and up to 4 alarm numbers can be displayed cyclically. The alarm number is displayed in the order in which warnings occur. The specific format is as follows:

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	Low pack volt	0	2	0~3	1	0		
2	Low cell volt	2	2	0~3	1	0		
3	High pack volt	4	2	0~3	1	0		
4	High cell volt	6	2	0~3	1	0		

5	Temp cell high charge	8	2	0~3	1	0		
6	Temp cell high discharge	10	2	0~3	1	0		
7	Temp cell low charge	12	2	0~3	1	0		
8	Temp cell low discharge	14	2	0~3	1	0		
9	Temp Relay high	16	2	0~3	1	0		
10	Over charge current	18	2	0~3	1	0		
11	Over discharge current	20	2	0~3	1	0		
12	Cell volt diff	22	2	0~3	1	0		
13	Low SOC	24	2	0~3	1	0		
14	reserve				1	0		
15	reserve				1	0		
16	reserve				1	0		

5.5 (BATT ST2) ID:0x18F128F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	Capacity Remain	0	16	0~1000	0.1	0	Ah	Capacity Remain
2	Rate Capacity	16	16	0~1000	0.1	0	Ah	Rate Capacity
3	Cycle count	32	16	0~60000	1	0		
4	SOH	48	8	0~100	1	0	%	

5.6 (Ctrl_INFO) ID:0x18F0F428 (Pending)

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	MaskCode	0	8					1: Allow control 0: Disable control bit0: Charging control bit1: Discharge control
2	ChgSw	8	8	0~1				Charging switch, 0: Off 1: On
3	DchgSw	16	8	0~1				DisCharging switch, 0: Off 1: On

5.7 (ChgRequest_INFO) ID:0x18F0F472

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	Batt Volt Request	0	16	0~10000	0.1	0	V	Maximum requested charging volt limit
2	Batt Curr Request	16	16	0~10000	0.1	0	A	= 95% Over current charge LV1

5.8 (BmsSwSta) ID: 0x18F528F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	Pre-discharge Status	0	1	0~1		0		Relay/ Mosfet Pre-discharge status 0: open 1: closed
2	Discharge Status	0	1	0~1		0		Relay/ Mosfet Discharge status 0: open 1: closed
3	Charge Status	0	1	0~1		0		Relay/ Mosfet Charge status 0: open 1: closed

5.9-17 (CELL_VOLT_FULL) ID: 0x18E028F4 - 0x18E728F4**CAN ID : 0x18E028F4**

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	CELL 1	0	16	0~10000	1	0	mV	
2	CELL 2	0	16	0~10000	1	0	mV	
3	CELL 3	0	16	0~10000	1	0	mV	
4	CELL 4	0	16	0~10000	1	0	mV	

0x18E028F4 AD 0E AB 0E A3 0E A6 0E

AD 0E : cell 1 3757mV

AB 0E : cell 2 3755mV

A3 0E : cell 3 3747mV

A6 0E : cell 4 3750mV

CAN ID : 0x18E128F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	CELL 5	0	16	0~10000	1	0	mV	
2	CELL 6	0	16	0~10000	1	0	mV	
3	CELL 7	0	16	0~10000	1	0	mV	
4	CELL 8	0	16	0~10000	1	0	mV	

CAN ID : 0x18E228F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	CELL 9	0	16	0~10000	1	0	mV	
2	CELL 10	0	16	0~10000	1	0	mV	
3	CELL 11	0	16	0~10000	1	0	mV	
4	CELL 12	0	16	0~10000	1	0	mV	

CAN ID : 0x18E328F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	CELL 13	0	16	0~10000	1	0	mV	
2	CELL 14	0	16	0~10000	1	0	mV	
3	CELL 15	0	16	0~10000	1	0	mV	
4	CELL 16	0	16	0~10000	1	0	mV	

CAN ID : 0x18E428F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	CELL 17	0	16	0~10000	1	0	mV	
2	CELL 18	0	16	0~10000	1	0	mV	
3	CELL 19	0	16	0~10000	1	0	mV	
4	CELL 20	0	16	0~10000	1	0	mV	

CAN ID : 0x18E528F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	CELL 21	0	16	0~10000	1	0	mV	
2	CELL 22	0	16	0~10000	1	0	mV	
3	CELL 23	0	16	0~10000	1	0	mV	
4	CELL 24	0	16	0~10000	1	0	mV	

CAN ID : 0x18E628F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	CELL 25	0	16	0~10000	1	0	mV	
2	CELL 26	0	16	0~10000	1	0	mV	
3	CELL 27	0	16	0~10000	1	0	mV	
4	CELL 28	0	16	0~10000	1	0	mV	

CAN ID : 0x18E728F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	CELL 29	0	16	0~10000	1	0	mV	
2	CELL 30	0	16	0~10000	1	0	mV	
3	CELL 31	0	16	0~10000	1	0	mV	
4	CELL 32	0	16	0~10000	1	0	mV	

5.18 (CELL_TEMP_FULL) ID: 0x18F228F4

No	Parameter	Start position	bit length	scope	Resolution	Offset	unit	Remark
1	Temp Relay	0	8	-50~200	1	-50	°C	
1	Temp Shunt	8	8	-50~200	1	-50	°C	
1	Cell Temp1	16	8	-50~200	1	-50	°C	
1	Cell Temp2	24	8	-50~200	1	-50	°C	
1	Cell Temp3	32	8	-50~200	1	-50	°C	
1	Cell Temp4	40	8	-50~200	1	-50	°C	
1	Cell Temp5	48	8	-50~200	1	-50	°C	
1	Cell Temp6	56	8	-50~200	1	-50	°C	